

An optimality treatment of syntactic inversions in English verse

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Abstract

This paper looks at syntactic inversions, a phenomenon first brought to the attention of Generative Metrics in Youmans [Youmans, G., 1983. Generative tests for generative meter. *Language* 59, 67–92]. These inversions are poetic lines that show disruption to the syntax so as to better satisfy the metrical constraints. In this paper, I show that two metrical constraints force these inversions, and that lines surface even with the violation of these metrical constraints. Furthermore, I argue that the inversions are best modeled by interleaving syntactic and metrical constraints, as certain types of syntactic constructions never occur. This suggests that some syntactic constraints outrank metrical ones, while other syntactic constraints are themselves outranked by the metrical constraints.

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0. Introduction

Poetic meter offers an excellent opportunity to examine the interaction of multiple domains of grammar simultaneously. Lines of poetry may be characterized in terms of prosodic, syntactic, artistic and semantic well-formedness. However, it is not the case that all considerations can be simultaneously satisfied in every line of poetry. An example of how syntax and prosody conflict comes from Youmans (1983), whose seminal work in *Generative Metrics* showed that syntactically ill-formed lines serve to increase metrical well-formedness. Compare the syntax in the two lines in (1). The first example, (1a) is from

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Hamlet (5.1.223), while the second example, (1b), is a construct from Youmans (1983, p. 67). The construct line places the stressed syllables in *weakest* and *bodies* into weak positions, while the actual line avoids such placements for the stresses in polysyllabic words. The scansion of the iambic pentameter lines below indicates stressed syllables with an “x” and unstressed syllables with a “.”, while the template corresponds to the abstract weak and strong metrical positions.

(1) a. SCANSION	.	x	.		x	.		x	.		x	.		x
LINE	Con	céit	in		wéake	st	bódies	stróngest		wórks				
TEMPLATE	W	S	W		S	W	S	W		S	W		S	
b.														
SCANSION	.	x		x		x	.		.		x	.		x
LINE	⇐	Con	céit	<u>wórks</u>	stróngest		in	<u>wéake</u> st	<u>bódies</u> .					
TEMPLATE		W	S	W		S	W	S		W	S		W	S

The two lines can be compared in terms of metrical and syntactic well-formedness. Metrically, the line in (1a) is better than that in (1b) because it is more rhythmic and because there are no stresses that align to weak positions in the template. Syntactically, the opposite is true, because (1b) follows the canonical word order typical of English sentences. In contrast, (1a) has the lexical verb falling in utterance-final position two constituents away from the subject. The disruption of canonical syntax in poetic verse is termed a *syntactic inversion* by Youmans. The importance of syntactic inversions to Generative Metrics was first shown in Youmans (1982), with later work including Youmans (1983, 1989, 1996) and Youmans and Li (2003). Youmans’ work showed that the vast majority of syntactic inversions serve to improve Shakespeare’s meter. In particular, syntactic inversions support the claim that there is a constraint on the meter that restricts the stresses in polysyllables from filling weak positions in the meter, known as Kiparsky’s Monosyllabic Word Constraint (Kiparsky, 1975, 1977).

This paper examines the meter of Shakespeare’s sonnets with a particular focus on the syntactic inversions. The approach taken here is the modeling of syntactic inversions in an Optimality Theoretic treatment. The poetic grammar requires that prosodic constraints dominate syntactic ones (see also Golston, 1995; Fitzgerald, 1994, 1995, 1998, 2001, 2003; Rice, 1997a,b; Golston and Riad, 2000). The schema for this type of ranking appears in (2):

(2) Constraint Ranking Schema for Poetry

METRICAL CONSTRAINTS $\gg$ SYNTACTIC CONSTRAINTS

This paper offers an Optimality analysis of syntactic inversions in Shakespeare’s verse, drawing upon the basic model of the poetic grammar proposed in (2). There are three basic findings that emerge from this analysis. First, I show that only certain types of syntactic inversions occur. Second, I show that two metrical constraints are the driving force behind inversions, only one of which is the Monosyllabic Word Constraint. Third, I show evidence that the Monosyllabic Word Constraint is itself violated, although it appears that

¹ Constructed lines of iambic pentameter are always set off with the ⇐ symbol to differentiate these from actual lines written by Shakespeare. Underlining in poetic lines indicates mismatches in the meter, where stressed syllables appear in weak metrical positions.

this only happens in order to satisfy one of the other metrical constraints on the poetry. The analysis presented here partially supports (2), but presents it as an incomplete schema. Instead, syntactic and metrical constraints must be interleaved, as argued in Fitzgerald (1994). A strict constraint ranking like (2) will otherwise incorrectly predict that any type of syntactic inversion can occur to improve the meter. Such results are not seen.

This paper is organized in the following way. The first section sets out the approach of Generative Meter and the importance of Youmans' original work on syntactic inversions in Shakespeare's verse. The second section presents an account of the attested inversions that is couched in terms of Optimality Theory and addresses the implications of this for the poetic grammar. The third section concludes the paper.

1. Generative meter and syntactic inversions

Meter is a repeating pattern, generally of a rhythmic nature. The repeating pattern can involve stressed and unstressed syllables, tone, syllable weight or quantity, or some combination of these qualities. Poetic meter is interesting from the linguistic perspective because it represents an abstract structure, and a set of behaviors that are acceptable deviation from that abstract pattern. Generative Meter is the quest to formulate iambic pentameter abstractly enough such that all rhythmic variations in actual performance in fact have an identical rhythmic pattern if examined at that level (Fabb, 1997). Generative Metrics uses linguistic representations, rules, and knowledge to determine when and how the poet deviates from the pattern.

The generative approach to meter reflects the early use of constraints in a linguistic approach. The components in this approach typically include a metrical template, matching rules, and a set of filters. The metrical template is an abstract sequence of metrical positions that are matched with phonological constituents in a given line of poetry; the matching rules govern how actual phonological material is matched to the metrical positions in specific lines. The filters are a set of conditions under which the matching rules do not hold.

Classical English iambic pentameter consists of a metrical template of ten abstract metrical positions of alternating strength (weak, 'W', and strong, 'S'). The metrical template has typically been characterized as in (3a),² with (3b) as an example of one way the template can be filled with actual unstressed and stressed syllables.

- (3)
- | | | | | | | | | | | |
|----|-----|------|-------|-----------|------|-----|-----|---|---|---|
| a. | W | S | W | S | W | S | W | S | W | S |
| b. | . | x. | x. | x | . | x | . | x | | |
| | The | lŷon | dýing | thrústeth | fóρθ | his | páw | | | |
| | w | sw | sw | s w | s | w | s | | | |

The matching rules allow any syllable to fill a strong position, but there are major restrictions on stressed syllables in weak positions. Kiparsky (1975, 1977) noted that stressed syllables only appear in weak positions if the stress is on a monosyllabic word, or if the stressed syllable is preceded by a phrase boundary. Actual lines from Shakespeare

² For a somewhat different conception of iambic pentameter, see Hanson and Kiparsky (1996) and Golston (1999).

and other poets who used iambic pentameter provide the evidence for the allowable range of possible instantiations of the template in (3a). Kiparsky's statement on the distribution of stressed syllables in iambic pentameter appears in (4):

(4) Monosyllabic Word Constraint: (Kiparsky, 1975, p. 583; 1977, p. 195)

Stressed syllables are prohibited in odd positions unless

- (a) the stress is on a monosyllabic word or
- (b) it is preceded by a phrase boundary

The first condition of this constraint can be seen in the two lines in (5). In the first line, *eruption* has a stressless initial syllable, but the second syllable is stressed. There is only one possible alignment to place the stress in a strong position in the meter; the word must start on a weak position in the metrical template. In the second line, the word *apparition* appears. Given the presence of stress on the initial and third syllables, there is only one possible way to align it to the meter so that the both stresses fall in strong position. The initial syllable must appear in a strong position.

- (5) a. . x . x . x . . . x
 This bôdes some strange erúption to our státe. (Ham 1.1.81)
 w s w s w s w s w s
- b. . . . x . x . x . x
 That if agáin this àpparítion côme, (Ham 1.1.35)
 w s w s w s w s w s

In addition to the placement of stresses in polysyllabic words, these lines illustrate several additional points. Most strong positions in the meter are relatively flexible in the type of syllables that they license, although there is a tendency for stressed syllables to appear in these positions. Tarlinskaja (1976) notes that Shakespeare places stressed syllables in the fourth and tenth positions in the meter (both strong) over 90% of the time. Weak positions are most likely filled by unstressed syllables, but stressed syllables are also possible. The examples in (5) also demonstrate that sequences of two or more unstressed syllables, known as a lapse, can occur; further below, there are examples showing the occurrence of two or more stressed syllables, known as a clash.³

The active role of the abstract poetic meter and constraints on the meter is illustrated by syntactic inversions in English meter. Youmans (1982, 1983) scanned the poetic lines from *Hamlet* and the *Sonnets*. He also rewrote the verse of Hamlet into normal prose order, and then compared poetic order with prose order. The proposal made by Youmans (1983) is that lines with more marked syntactic order correlate with various factors; his focus is on cases where the meter improves by changing the canonical word order. He tested Kipar-

³ There are two additional issues that are important to iambic pentameter: extrametrical syllables and lexical inversion. Extrametrical syllables are those syllables that are systematically not projected from the line, as in the extensive discussion in Fabb (2002). These are syllables of a particular phonological structure (for example, a syllabic sonorant). Lexical inversion is the second part of the Monosyllabic Word Constraint, which allows stresses in polysyllabic words to appear in a weak position at the beginning (or following the end) of a major phrase boundary. This has a major effect on line beginnings, which are permitted to begin with a stressed syllable.

sky's metrical constraint against an alternative metrical constraint conceived by Halle and Keyser (1971). The results favored Kiparsky's Monosyllabic Word Constraint. Example (6) illustrates this by comparing an actual Shakespearean line with its normalized prose counterpart.

(6) Syntactic Inversions (Son 56.2)

a. SCANSION	.	x	.	x	.	x	.	x	.	x
LINE	Thy	édge	should	blúnter	bé	than	áppetíte			
TEMPLATE	W	S	W	S	W	S	W	S	W	S
b. SCANSION	.	x	.	x	x	.	.	x	.	x
LINE	←Thy	édge	should	bé	<u>blúnter</u>	than	áppetíte			
TEMPLATE	W	S	W	S	<u>W</u>	S	W	S	W	S

The two lines show different ways to satisfy conflicting constraints on the syntax and the meter. The line in (6a) is metrically well-formed because none of the stresses in polysyllabic words appear in weak positions. It also displays an alternating rhythm where there are neither stresses that clash nor excessive sequences of unstressed syllables. However, the syntax of (6a) is disrupted; the verbal complement has moved leftward to come immediately before the verb. The canonical word order for this line is shown in (6b), but this line is a constructed example. In (6b), the stressed syllable in *blunter* falls in a weak position, a violation of the Monosyllabic Word Constraint. The example in (6a) is termed a **syntactic inversion**: the word order changes from typical to less typical (or even atypical). The comparison between (6a) and (6b) demonstrates that syntactic inversions can improve the meter.

In fact, three major forces motivate syntactic inversions in English: emphasis, rhyme, and the metrical constraints that govern the distribution of stressed and unstressed syllables. The focus of this paper lies in metrically-driven syntactic inversions, based on the seminal work by Youmans (1982, 1983, 1989). Youmans' work examines syntactic inversions in *Hamlet* and in Shakespeare's *Sonnets*; the inclusion of *Hamlet* is interesting because the work contains poetry and prose sections (as well as sections that are rhymed and sections that are blank verse).

The claim made by Youmans (1983, 1989, 1996) is that there is a strong positive correlation between syntactic inversions and metrical principles. In other words, syntactic inversions let the poet better satisfy the constraints on the meter. The overwhelming majority of inversions that are not driven by rhyme create a positive metrical effect. The chart in (7) shows the percentage of lines with non-rhyme syntactic inversions in the three different sources.

(7) Non-rhyme Syntactic Inversions (Youmans, 1983, p. 75)

Data Source	# of Lines	# of inversions	Percent
Prose in <i>Hamlet</i>	c. 1035	24	2.3
Verse in <i>Hamlet</i>	2751	351 (exclusive of rhyme)	12.8
<i>Sonnets</i>	2155	152 (exclusive of rhyme)	7.1

Of the 152 poetic inversions in the *Sonnets*, 110 create a positive metrical effect. Nearly three-fourths of these non-rhyme inversions have a positive metrical effect. In *Hamlet*, more than three-fourths of the non-rhyme syntactic inversions have a positive metrical effect (279 out of 351 total lines). Overall, only 15 lines in both works create a negative metrical effect; the remaining inversions create a minor or no effect in terms of the metrical pattern.

Non-rhyme syntactic inversions do not display a pervasive disruption of any type of constituency. Rather, they can be classified in terms of their syntactic effects. Three basic types of syntactic structures allow inversion: pronominal adjectives and the nouns they modify (e.g., *brown donkey*), verbs and complements (e.g., *buying cars*), and verbal adjuncts or adverbs (*dance in the moonlight*).

The use of inversion to switch the canonical adjective–noun order into a noun–adjective sequence was common enough to have been included as instructions to poets by George Gascoigne in 1575 (Gascoigne, 1868). Here are his words on the subject:

You shall do very well to use your verse after the English phrase, and not after the manner of other languages ... if we should say in English a woman fair, a house high, etc. it would have but small grace: for we say a good man, and not a man good, etc.... Therefore even as I have advised you to place all words in their natural or most common and usual pronunciation, so would I wish you to frame all sentences in their mother phrase and proper Idioma, and yet sometimes the contrary may be borne, but that is rather where rhyme enforceth, or per licentiam Poëticam, than it is otherwise lawful or commendable. (Gascoigne, 1868, p. 37)

The inversion of an adjective noun sequence most often occurs to satisfy one or more offenses to the Monosyllabic Word Constraint, as the pair in (8) demonstrates. The canonical order where the adjective precedes the noun, as in (8b), misplaces the stress in the polysyllable *impregnable*. This polysyllable is more appropriately aligned to the meter in (8a) by reversing the order of the adjective and the noun.

(8) Inversion of an adjective–noun sequence (Son 65.7)

- a.
- | | | | | | | | | | |
|------|-------|-------------|-----|-----|----|-------|---|---|---|
| . | x | . | x | . | . | . | . | . | x |
| When | rócks | impregnable | are | not | so | stóut | | | |
| W | S | W S W S | W | S | W | S | | | |
- b.
- | | | | | | | | | | |
|--------|-------------|-------|-----|-----|----|-------|---|---|---|
| . | . | x | . | . | x | . | . | . | x |
| ← When | impregnable | rócks | are | not | so | stóut | | | |
| W | S W S W | S | W | S | W | S | | | |

Verbal complements, such as direct objects or subject predicates, also become preposed with some frequency.⁴ The example in (9) illustrates how such inversion operates in order to place a disyllable with a trochaic stress pattern in a strong–weak alignment to the meter (9a), instead of aligning weak–strong (9b). The example above in (6) is the same type, but with a subject predicate acting as the inverted complement.

⁴ My thanks to Gilbert Youmans for providing examples in this section.

(9) Inversion of a direct object (Ham 1.3.50)

- a. . x . x . x . x . x
Himsélf the primrose páth of dálliance tréads
W S W S W S W S W S
- b. . x x . x . x . x .
⇐ Himsélf tréads the primrose páth of dálliance
W S W S W S W S W S

Another occurring pattern is the inversion of adjuncts, optional elements in the verb phrase like adverbs or prepositional phrases acting like adverbials. These examples are often somewhat messier, as in this example repeated from above.

(10) Inversion of an adverbial (Ham 5.1.223)

- a. . x . x . x . x . x
Concéit in wéakest bódies strongest works
W S W S W S W S W S
- b. . x x x . . x . x .
⇐ Concéit works strongest in wéakest bódies.
W S W S W S W S W S

The generalization suggested by these last few examples is that inversion is more likely to occur if the structure that would be disrupted is a head and its complement or a head and its modifier or adjunct. English canonical word order places the head before the complement, but after the specifier. In contrast, we rarely find that an inversion results in a specifier being preceded by its head, such as a determiner following a noun, or a degree word like *very* following an adjective or adverb. Specifier-head ordering appears to resist the pressure to invert, while the relationship between heads and complements is more amenable to syntactic inversion. The inversion of subject and verb can occur, as the example in (11) illustrates, although such examples occur infrequently. The inversion in this example may be motivated by one of two possible causes, either the avoidance of stress clash in *thoughts die*, or the placement of the clitic sequence *thy thoughts* according to the Monosyllabic Word Constraint.

(11) Subject-Verb Inversion (Ham 3.2.214)

- a. . x . x . . x x . x
But díe thy thoughts when thy fírst lórd is déad.
W S W S W S W S W S
- b. . . x x . . x x . x
⇐ But thy thoughts díe when thy fírst lórd is déad.
W S W S W S W S W S

One point of interest is that the subject-verb construction, like the head-complement construction, is subject to leftward movement in non-poetic syntax. So for example, questions formation in English involves subject–auxiliary inversion, and processes of topicalization involve leftward dislocation of postverbal constituents. But the constituent composed of a specifier and its head is generally not subject to the same movement processes. The prevalence of movement in certain syntactic constructions and the (near) absence in other syntactic constructions suggest that the proper schema for constraint ranking in poetry is more appropriately characterized as in (12):

(12) Constraint Ranking Schema for Poetry
SYNTACTIC CONSTRAINTS $\gg$ METRICAL CONSTRAINTS $\gg$ SYNTACTIC CONSTRAINTS

One way to envision this grammar is that some constraint requiring functional categories to precede lexical categories in English (so *the dog* rather than **dog the*) ranks above the Monosyllabic Word Constraint, thus preventing inversions in such cases.⁵ However, the constraint requires that heads precede their complements. For this paper, I adopt three basic syntactic constraints in a simplified account of the syntax. These constraints appear in (13). The first two are alignment constraints from the literature on Optimality Syntax; the third constraint is one proposed here to deal with the adjective–noun ordering in a straightforward fashion, although the issues surrounding this are more complex.⁶ In (13d), the interaction of the first two constraints is shown to generate the order appropriate for English syntax.

- (13)
- a. HEADLEFT (HdLFT)
Every X-zero is at the left edge of an X-max (Grimshaw, 1997)
 - b. SPECLFT
Every specifier is at the left edge of an X-max (Grimshaw, 1997)
 - c. ADJ-N
Adjectives precede the nouns that they modify.
 - d. Word order when SPECLFT $\gg$ HEADLEFT⁷

	SPECLFT	HdLFT
a.  [Spec Head Comp]		*
b. [Spec Comp H]		**!
c. [H Comp Spec]	*!*	
d. [Comp H Spec]	*!*	

⁵ Gilbert Youmans (personal communication) points out that one poet who is an exception to reordering articles and nouns is E.E. Cummings.
⁶ Note that this paper is putting aside the issue of adverbials, but the model presented in this paper should extend to these types of inversions without a problem.
⁷ As the example tableau shows, the optimal placement of the head will generate a violation of the constraint if there is a specifier. In order to clarify the focus onto the constructions involved in syntactic inversions, I will only mark violations related to the target construction, when the target construction involves a specifier.

In the next section of the paper, I turn to an account of how these three constraints interact with metrical constraints, particularly the constraint that regulates the positioning of stresses in polysyllabic words.

2. Modeling syntactic inversions

This next section sketches out a highly simplified Optimality Theoretic grammar of Shakespearean meter to predict meter-driven syntactic inversions, in part because the focus of this paper is the syntactic inversions.⁸ The syntactic constraints in (13) will interact with a constraint that regulates the alignment of stressed syllables with the metrical template, MATCH STRESS. MATCH STRESS prohibits stress in polysyllabic words from appearing in weak positions in the meter. The statement of this constraint comes from work on quatrain structure in folk songs by Hayes (2000) and Hayes and MacEachern (1998).⁹ It thus replicates the first half of Kiparsky's Monosyllabic Word Constraint. Recall that this paper will not deal with lexical inversions, the subject of the second half of that constraint.

(14) MATCH STRESS

Assess a violation if:

- σ_i and σ_j (in either order) are linked to grid positions G_i and G_j respectively;
- σ_i has stronger stress than σ_j ;
- G_j is stronger than G_i ; and
- σ_i and σ_j occupy the same simplex word.

The four constraints presented so far will instantiate the interleaving of metrical and syntactic constraints as in (15):

(15) Constraint Ranking for Syntactic Inversions

SPECLFT $\gg$ MATCH STRESS $\gg$ HDLFT, ADJ-N

This ranking predicts that syntactic inversions will occur to place the stresses in polysyllabic words into strong positions of the meter, but such inversions can only affect the orderings of adjective–noun sequences or heads (specifically, verbs) and their complements (direct objects, subject predicates).

Now let us evaluate an unrhymed line from *Hamlet* that allows a syntactic inversion to occur. Example (16a) shows *mórtal* in strong–weak position, while (16b) places it into weak–strong position. The constructed line is (16b), which reflects a satisfaction of syntactic well-formedness; (16a) displays a syntactic inversion, where the noun precedes the adjective.

⁸ For more on Optimality Theory, see Prince and Smolensky (1993), among many others. For more on treatments of meter within Optimality Theory, see Fitzgerald (1995, 1999, 2001, 2002, 2003, 2006), Golston (1998), Golston and Riad (2000), Hanson and Kiparsky (1996), Hayes and MacEachern (1998), Hayes (2000).

⁹ As written, a stressed monosyllable will not violate this constraint. However, a revision to the notion “simplex word” to include the prosodic word (a lexical word together with its unstressed clitics) would regulate the distribution of sequences like *the dog* or *dreamt of*. Kiparsky (1977) argues that there is a tendency for such sequences to align with the lexical stresses in strong position. The grouping of such sequences is reminiscent of the behavior of clitic groups in Hayes (1989).

(16)	a.	. x x x . x . . . x	
		Unléss <u>thíngs</u> mórtal móve them not at áll,	
		W S W S W S W S W S	
	b.	⇐ . x x . x x . . .	
		Unléss <u>mórtal</u> <u>thíngs</u> móve them not at áll,	
		W S W S W S W S W S	

The tableau in (17) compares these two in terms of their satisfaction of the constraints. The actual line, (17a), violates the syntactic constraint, ADJ-N. However, it performs better on the metrical constraint, MATCH STRESS, than the non-inverted line does in (17b). As Youmans (1983) observes, the syntactic inversion improves the metrical profile of the line. The syntactic violation removes (or reduces, in some cases) the metrical violation.

(17) Evaluation of competing rhythmic and syntactic patterns (Ham 2.2.516)

	SPECLFT	MATCH	HdLFT	ADJ-N
a). x x x . x . . . x ☞ Unléss thínks mórtal móve them not at áll W S W S W S W S W S				*
b). x x . x x . . . x Unléss mórtal thínks móve them not at áll W S W S W S W S W S		*!		

This next example compares another pair of lines, with the actual line from *Hamlet* exhibiting a syntactic inversion in which the object precedes the lexical verb. Such an inversion will violate HEADLEFT in order to satisfy MATCH STRESS. MATCH STRESS has two violations because the repositioned adjective–noun sequence consists of two disyllables with initial stress. Shifting that constituent to the left of the lexical verb results in a final stress and perfectly aligned stresses in the polysyllabic words.

(18) Syntactic inversion (Ham 5.2.41)¹⁰

	SPECLFT	MATCH	HdLFT	ADJ-N
a). x . x . x . x . x ☞ As péaceshóuld stíll her w héaten gárlánd wéar w s w s w s w s w s			**	
b). x . x x . x . x . As péace shóuld stíll wéar her w héaten gárlánd w s w s w s w s w s		***	*	

With MATCH STRESS ranked above both HEADLEFT and ADJ-N, the types of inversions in (17) and (18) are predicted. While two of the syntactic constraints can surface as violated in the optimal verse line, this option is not available when it comes to constraints on the location of specifiers. This is shown in the comparison below, which compares different candidates for a rhymed line from the *Sonnets*. The final rhyme of the line is not an issue,

¹⁰ The specifier *her* triggers one violation on HEADLEFT for both candidates; the additional violation for (18a) relates to the inversion.

as each candidate ends with the same word, *stout*. The first candidate disrupts the canonical ordering of adjectives and nouns, and so violates the ADJ-N constraint. Candidate (19b) maintains the canonical order, with the result that the placement of *imprégnable* triggers a violation on MATCH STRESS. The third candidate allows the proper placement of stress in the polysyllable, but does so by shifting the complementizer out of the specifier position. This triggers a fatal violation of SPECLEFT, because the complementizer does not appear to the left of the clause.

(19) Interleaving of Metrical and Syntactic Constraints (Son 65.7)

	SPECLEFT	MATCH	HDLEFT	ADJ-N
a) . x . x x When rôcks imprégnable are not so stóut W S W S W S W S W S				*
b) . . x . . x . . x When imprégnable rôcks are not so stóut W S W S W S W S W S		*!		
c) . x . . x . . . x Imprégnable rôcks are when not so stóut W S W S W S W S W S	*!			

These tables above have demonstrated that the avoidance of MATCH violations triggers inversions that correspond to the syntactic constraints ranked below MATCH. However, MATCH is not the only metrical constraint that triggers syntactic inversions. The actual and construct lines in (20) give evidence for an inversion of the subject and lexical verb. However, as the construct line shows in (20b), there are no polysyllabic words with misplaced stresses. So the inversion cannot be related to a violation on MATCH STRESS. A comparison of the two lines shows that the subject–verb inversion results in a more rhythmic profile to the line. In other words, the inversion avoids two stressed syllables.

- (20) a. . x . x . x . x . x
Then lóok I déath my dáy's should éxpíàte
W S W S W S W S W S
- b. . . x x . x . x . x
⇐ Then I lóok déath my dáy's should éxpíàte
W S W S W S W S W S

The avoidance of stress clash has been shown as a factor in various phonological and syntactic processes. The constraint that typically restricts stress clash is *CLASH.¹¹

- (21) *CLASH (Nespor and Vogel, 1979; Nespor and Vogel, 1989; Prince, 1983; Selkirk, 1984)

No sequence of adjacent stressed syllables.

In order to force an inversion, *CLASH must outrank the syntactic constraint HEADLEFT. The tables in (22) motivates this constraint ranking. Candidate (22a) displays a direct

¹¹ For help with examples relevant to *CLASH, my thanks to Gilbert Youmans.

object that has been moved to the left of the clause boundary, leading to a violation of HEADLEFT. The line also violates MATCH, with the misplaced stress in *barren*, and *CLASH. In (22b), the direct object occurs postverbally, so there is no violation of HEADLEFT. However, there are two violations of *CLASH. The stress in *barren* is still misplaced, so this line also has a violation of MATCH. If the rankings between *CLASH and HEADLEFT were reversed, (22b) would be incorrectly selected as optimal because it would then satisfy the higher ranked HEADLEFT.

(22) *CLASH ≫ HEADLEFT (Son 12.5)

	SPECLFT	MATCH	*CLASH	HDLFT	ADJ-N
a) . x . x . x x . . x ☞ When lófty trées I sée bárrén of léaves W S W S W S W S W S		*	*	*	
b) . . x x . x x . . x When I sée lófty trées bárrén of léaves W S W S W S W S W S		*	**!		

The relationship of *CLASH to inversion can also be seen in more complicated examples that involve the inversion of subject and verb, such as appear in (23) and (24). Under Grimshaw’s model of OT syntax, question-driven inversion of the subject and verb do not result in violations of SPECLFT. The movement of the head may result in a violation of HEADLEFT, depending on the particular configuration. In (23a), the verb precedes the subject, which violates HEADLEFT. The metrical constraints are satisfied by this candidate. In (23b), there is also a violation of the syntactic constraint because the subject is the specifier and causes the misalignment of the verb with the left edge. However, this line results in stress clash, which is avoided in the inverted line of (23a).

(23) *CLASH ≫ HEADLEFT (Son 22.4)

	SPECLFT	MATCH	*CLASH	HDLFT	ADJ-N
a) . x . x . x . x . x ☞ Then lóok I déath my dáy’s should éxpiàte W S W S W S W S W S				*	
b) . . x x . x . x . x Then I lóok déath my dáy’s should éxpiàte W S W S W S W S W S			*!	*	

These examples have shown that two metrical constraints play a role in forcing syntactic inversions, MATCH and *CLASH. One interesting by-product in table (22) above was the result that the actual line surfaces even with a violation of MATCH. In the next tableau, I provide an additional example that MATCH can be violated in actual attested lines. I suggest that this is due to the fact that the inversions still result in a metrical improvement. In other words, the attested lines with MATCH violations perform better on the constraints than their non-inverted counterparts.

The crucial difference between (24a) and (24b) relates to the performance on MATCH STRESS. The attested line only misplaces the stresses in *borrower* and *lender*, while (24b) misplaces only *neither*. The shift of the copular verb to the final stressed position allows everything to shift leftward by one position, resulting in a better scansion.

(24) Violation of MATCH in the actual line (Ham 1.3.76)

	SPECLFT	MATCH	*CLASH	HDLFT	ADJ-N
a) x . . x . . . x . x Néithér a bórrower nor a lénder bé w s w s w s w s w s		*		*	
b) x x . . x . . . x . Bé néithér a bórrower nor a lénder w s w s w s w s w s		**!	*		

The finding that optimal lines surface with violations of a highly-ranked constraint is exactly what we should expect under Optimality Theory. Constraint violability is one of the hallmarks of this theory. Thus a single violation of MATCH is always more optimal than additional violations on the same constraint, provided the performance stays constant on the constraints ranked above MATCH.

3. Conclusion

The pre-OT formalism of metrics is one that relies on constraints and filters. One exciting result that has emerged from Optimality Theory is the way that constraint ranking can be used to characterize the interaction of different constraints, and even different aspects of grammar such as syntax and phonology. This paper has tried to shed some light on how metrical and syntactic constraints interact in poetic grammar. The work of Youmans (1983) provides a straightforward testing ground, especially given that the original research was designed to determine the nature of the metrical constraint or constraints forced the inversions.

The implementation of Youmans' findings in Optimality Theory has led to three results. First, there is a restriction on what types of syntactic inversions can occur. The overwhelming tendency is for inversions to involve adjective–noun sequences or verbs and verbal complements. Head-complement inversion is more likely than specifier-head inversion, which only occurs in the case of subjects and their verbs. This asymmetry, where subject–verb inversion is the only type of specifier-head inversion, is unsurprising given the fact that there has been considerable research into subject movement in syntactic theory. A second finding is that at least two different metrical constraints force syntactic inversions, MATCH STRESS and *CLASH. A third finding is a refinement to the conception of poetic grammar, arguing that it involves the interleaving of syntactic and metrical constraints. A number of researchers have argued for constraint ranking in which syntactic constraints can be at least partially dominated by prosodic constraints (Fitzgerald, 1994, 1995, 2001; Golston, 1995; Rice, 1997a; Rice, 1997b; Golston and Riad, 2000). This is compatible with other analyses arguing that metrical constraints dominate syntactic ones; most such analyses examine a single syntactic constraint that reflects this ranking. This paper has shown that the split ranking, whereby metrical constraints are dominated by some syntactic constraints and dominate other syntactic constraints, correctly models the patterns that actually occur in syntactic inversions.

While this study has left out certain aspects of iambic pentameter, it has provided a preliminary characterization for how syntactic inversions fit into an Optimality Theoretic treatment of iambic pentameter. One extension for this study in the future lies in how

syntactic inversions interact with the alternative conceptions of iambic pentameter, such as [Hanson and Kiparsky \(1996\)](#). This avenue of research thus has the potential to shed light on the nature of constraint interaction and on the proper characterization of this classic English verse form.

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